

An Interactive Neural Network-Based System for Confined Stylization of Product Design

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Introduction

- Image Style Transfer is a computer vision technique to recompose the content of an image in the style of another. The problem of image style transfer via deep neural networks has received numerous contributions due to the recent rapid growth of machine learning. In 2015, Gatys et al. [1] introduced a revolutionary method to transfer an artistic visual style of a style image into a photorealistic image using a Convolutional Neural Network (CNN). The fundamental concept is to use the VGG convolutional network to segregate and reconstruct the content and style images. (Fig. 1). [1]'s work has served as a basis to support the succeeding research and development of image style transfer.

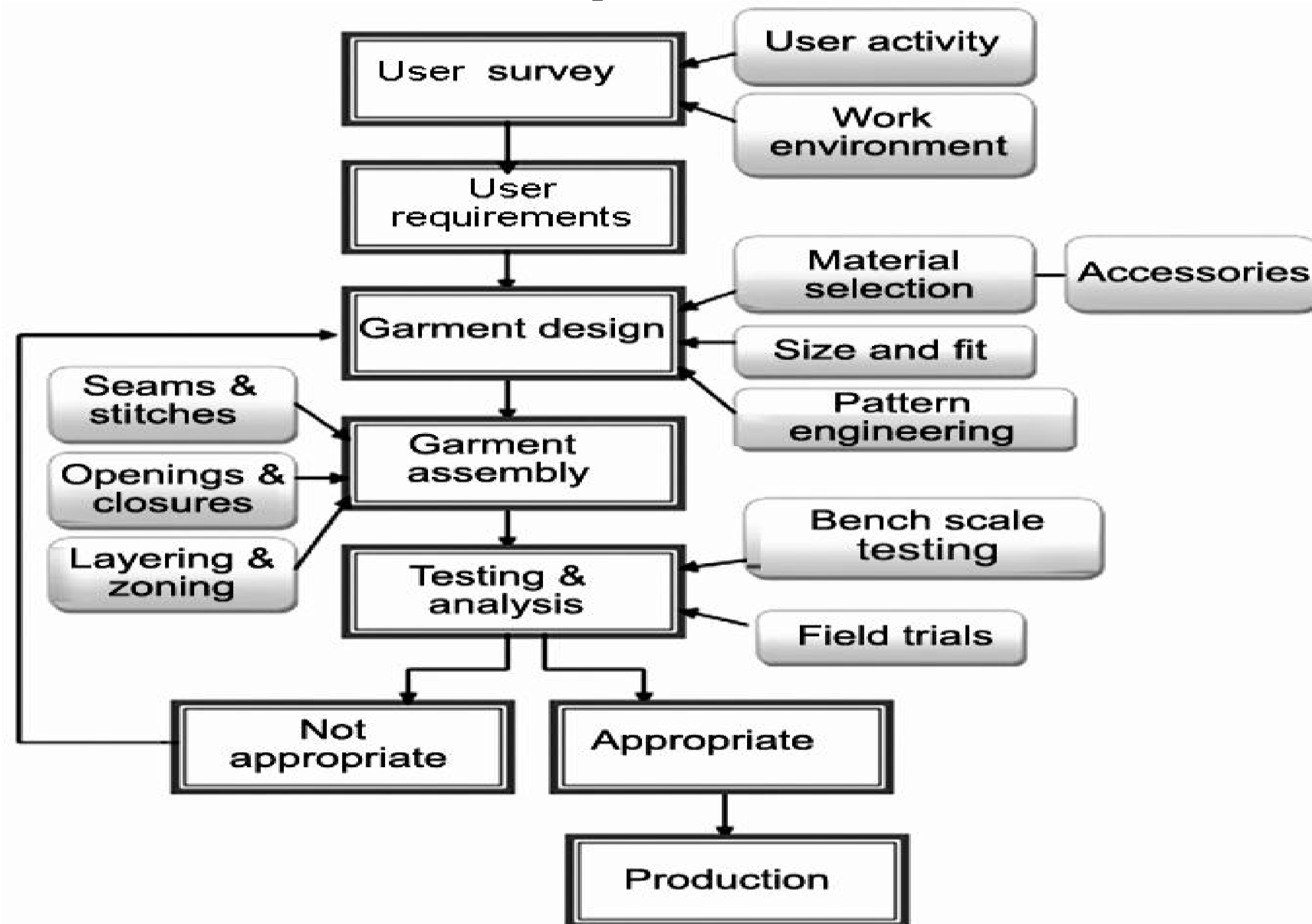
Fig. 1



- However, when directly applied to clothing style transfer, the present techniques do not explore much on enabling users to control the confined transfer region of an image, such as separating particular clothes like trousers or T-Shirts from a figure. To accomplish the goal, this research suggests an interactive image stylization technique. The Grab Cut Algorithm [2] was employed to extract additional image from the content image, the product designer can simply drag a rectangle around the desired product image. Then, a distance loss function was introduced to preserve the shape of the product during stylization. A GUI application that implements the suggested stylization strategy was created. The experimental result shows that the original clothing shape can be well reserved and our suggested approach is feasible to design product images.

- For this research, a GUI application has been developed and will be served as a fashion visualization tool for the professional designer during the initial garment design phase (Fig 2.). While there are already fashion visualization tools existing in the market like Digital Fashion Pro and CLO3D, the major difference between them and our application is obvious: our application can stylize the specific region of the clothes with arbitrary style image provided by designer while the common visualization tools can only provide their limited choices of color and pattern for clothes stylization. The stylized clothes which have another color, texture or pattern will be visualized by our application. As shown in Fig. 2., the fashion designer can determine whether it is feasible to proceed to the next phases by considering different factors, like the production cost of the stylized clothe, clothes' material selection, customer survey and requirements.

Fig. 2



Methods

1. Overall Procedure of Stylization

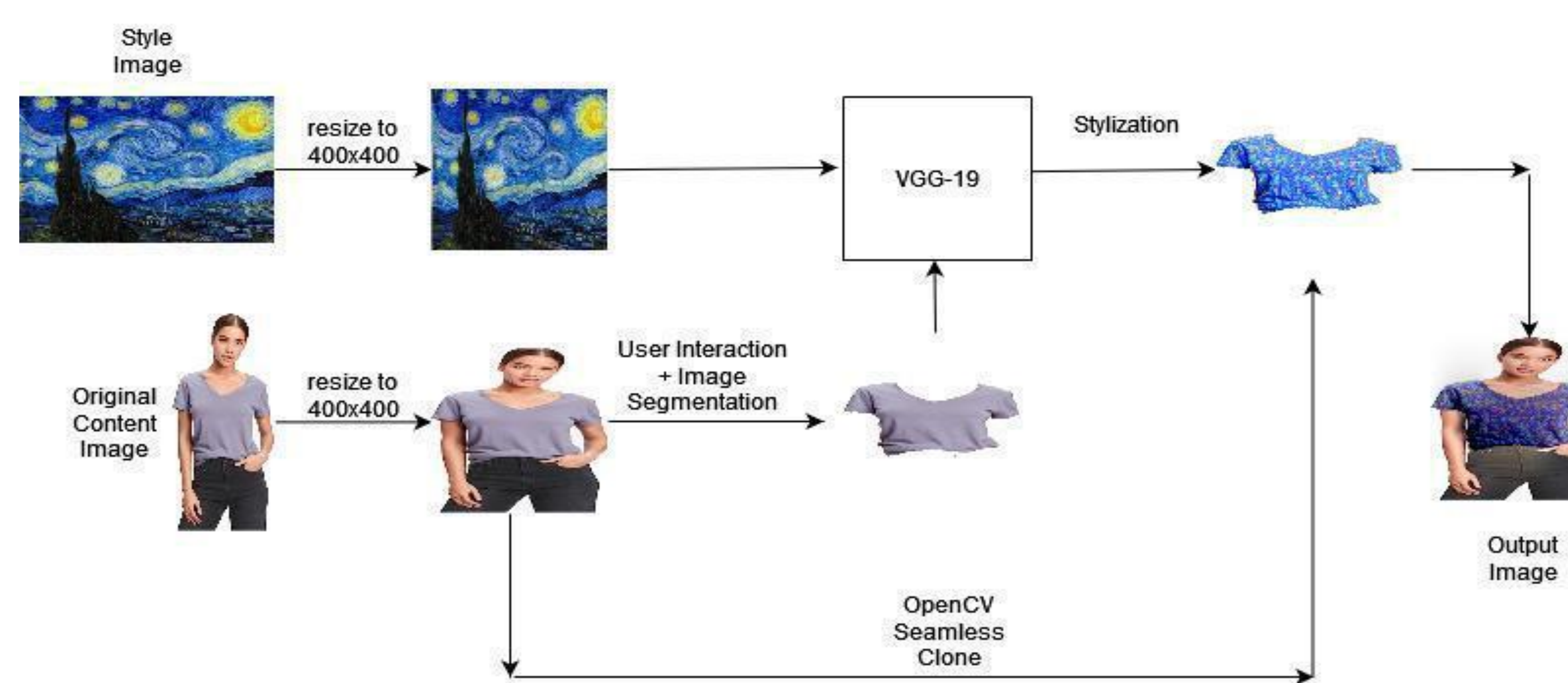
(Fig. 3) shows the overall process of image style transfer method as proposed in this paper. The proposed method was implemented by Python 3 and a GUI application was primarily written. The following operations will be performed by the application:

- (1) The dimension of the selected content image and style image will be resized into 400x400 to fit the layout of the GUI application.
- (2) The user will be required to drag a rectangle around the desired product of the content image to obtain an extracted image. This can be achieved by using GrabCut Algorithm [2].

(3) The features of the segmented image and style image will be extracted and stored using a pretrained VGG-19 network. Apart from the original content loss and style loss as proposed by [1], the additional distance transform loss [3] had also been introduced in this model for the calculation of total loss during neural style transfer. This will help to stylize the only pre-segmented region.

(4) The stylized segmented image will be cloned into the resized content image to generate an output image with non-altering background.

Fig. 3



2. GrabCut Algorithm

Based on the content image's color, grayscale and texture, it can be partitioned into multiple pieces called segments by Image Segmentation Algorithm. After segmentation, the features of the image exhibit disparities across diverse places yet show similarities in the same areas (Fig. 4). This study uses the GrabCut algorithm [2] to handle segmentation in order to give users the ability to precisely control the transfer position of an image.

Fig. 4



3. Stylization for Confined Region

The fundamental idea behind neural style transfer is to take the content and style information from the input image and combine them using a pretrained convolutional neural network (CNN). The input images used in this study include a content image with a complicated background, an extracted image that GrabCut Algorithm extracts from the content image, and a style image.

The VGGNet [1] generates a feature map from the filter responses to each layer for a given input image. An input image's content can be viewed as being represented by feature maps on particular layers. A feature space that is made to capture texture information is utilized to extract the style representation of an input image. It is made up of feature correlations that the Gram matrix provides in several layers. Given is the Gram matrix,

$$G_{ij}^l = \sum_k F_{ik}^l F_{jk}^l \quad (1)$$

where F_{ik}^l and F_{jk}^l refer to feature maps i and j in layer l .

For stylization, a content image P^* first inputs to the VGGNet and its feature maps in selected layer are stored as the content representation P_l on l th layer. Next, a style image a^* passes through the network. The sum of Gram matrices on every layer are computed and stored as style representation A_l of a style image. Then, the image to be generated, which is initialized as the content image, passes through the network. Using its feature maps, the content representation F_l and the style representation G_l of the generated image are computed on same layers as the respective representations. With content and style representations, loss functions used for generating images can be calculated. The content loss $L_{content}$ is calculated as shown in Eq. (2)

$$L_{content}(\vec{p}, \vec{x}, \vec{l}) = \frac{1}{2} \sum_{ij} (F_{ij}^l - P_{ij}^l)^2 \quad (2)$$

Style loss L_{style} can be calculated as shown in Eq. (3)

$$L_{style}(\vec{a}, \vec{x}) = \sum_{l=0}^L w_l E_l, \quad (3)$$

where

$$E_l = \frac{1}{4N_l^2 M_l^2} \sum_{ij} (G_{ij}^l - A_{ij}^l)^2 \quad (4)$$

w_l are weighting factors for the contribution of layer l to style loss, N_l is the number of filters in layer l , M_l is the dimensions of layer l . To stylize the confined region of the content image, [3] proposed a new loss function using distance transform of the input images. The distance transform assigns a value—the distance to the closest pixel that is silhouette—to each pixel in a binary image. The distance transform image D has same dimensions as original image, but its pixel values are the values of distance transform. For every pixel d_{ij} of the distance transform image D , emphasis with power n would look like,

$$d_{ij} = \begin{cases} 0 & \text{if inside of a silhouette} \\ d_{ij}^n & \text{otherwise} \end{cases} \quad (5)$$

The distance transform loss $L_{distance}$ would be

$$L_{distance} = \frac{1}{2} (\vec{p} \circ D_{content}^n - \vec{x} \circ D_{content}^n)^2 \quad (6)$$

By simply adding $L_{distance}$ to the total loss with weighting factor γ , we can obtain the following total loss function for model minimization:

$$L_{total} = \alpha L_{content} + \beta L_{style} + \gamma L_{distance} \quad (7)$$

Results



Fig. 5.: The results with zebra pattern and watercolour paintings as style images. (a) Style Image. (b) Content Image. (c) Extracted Image. (d) The method proposed by Gatys et al. [1] (e) Our Method

In Figure 5, we randomly picked four pictures from Internet as content images and style images. The shirts in Figure 5(c) was extracted by our application using the GrabCut Algorithm [2]. We compare our method with the stylization method proposed by Gatys et al. [1]. It was found that only our method keeps the background clean. In [1]'s method, the style from the style image is transferred to the whole content image rather than only clothes.

Conclusions and Future Directions

In this work, an interactive stylization approach was specifically suggested for product images. New cloth designs can be created by merging appropriate style photos, making it simpler for designer to create the product image as they want. The designer would determine to further proceed to the next design phase or production phase if the stylized product image was considered to be reasonable for manufacturing. Additionally, a GUI application had been primarily developed. Although the stylization method adopted by this paper was based on [1] and seems initial, the experimental results shows that it was still effective on creating fashionable cloth image. As there is currently not much existing research on confined region stylization, the stylization method used in this paper may be naïve. Given the simplicity of our approach, future researches may examine more sophisticated stylization algorithms, and experiments to compare them to other algorithms.

Examples of Literature Cited

1. L. A. Gatys, A. S. Ecker & M. Bethge, "A Neural Algorithm of Artistic Style," arXiv:1508.06576v2 [cs.CV], Sep. 2015.
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3. Atarsaikhan, Gantugs & Iwana, Brian & Uchida, Seiichi. (2018). Contained Neural Style Transfer for Decorated Logo Generation. 317-322. 10.1109/DAS.2018.78.